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Medicare Part D Plans Rarely Cover Brand-Name Drugs When Generics Are Available

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Abstract

Recent press reports and other evidence suggest that Medicare Part D plans may be encouraging the use of brand-name drugs instead of generics. However, the scope of such practices is unclear. We examined Medicare Part D formulary coverage and tier placement of matched pairs of brand-name drugs and generics to quantify how often preferred formulary placement of brand-name drugs is occurring within and across Part D plans and to assess the cost implications for Medicare and its beneficiaries. We found that in 2019, 84 percent of 4,176,772 Part D plan-product combinations had generic-only coverage (that is, the brand-name counterparts were not covered). Another 15 percent covered both the brand-name and generic versions of a product. For the small number of products whose brand-name versions were covered preferentially to their generic equivalents, beneficiary and Medicare prices were generally low for both products. Overall, we found that most Part D plan formularies are designed to encourage utilization of generics rather than their brand-name counterparts. Policy makers should continue to monitor Part D formulary coverage patterns to ensure consistent and generous coverage for generic drugs, given their important role in reducing prescription drug spending.

With forty-five million Medicare beneficiaries enrolled in Part D prescription drug plans and an estimated \$98 billion in program spending in 2019, formulary coverage decisions by Part D plan sponsors can affect overall retail prescription drug spending in the US.¹ Formulary coverage and tier placement—which drugs plans cover and how much patients pay for them, respectively—influence whether or not a particular drug is affordable and ultimately accessible to patients.

When brand-name drugs lose patent-protected status and generic drugs become available, utilization typically shifts to lower-cost generic drugs (encouraged through formulary placement and automatic pharmacy substitution laws, among other policies), which has traditionally generated substantial savings for patients and payers.^{2–5} However, recent press reports and other evidence suggest that Part D plans are increasingly listing certain generics on nonpreferred tiers (those with higher cost sharing),^{6,7} listing some generics less favorably than brand-name drugs,^{8,9} or not including certain generics on the formulary at all.^{6,7,9}

Stakeholders and scholars have offered numerous reasons for the observed shift of generics to nonpreferred tiers in Part D, including financial incentives associated with rebate arrangements, brokered by pharmacy benefit managers, that favor brand-name drugs over generics.^{10–13} Generally, rebates are negotiated for brand-name drugs but not for traditional generics. In Part D, these rebates may be used by plans to offset their total drug costs, which may help control premiums paid by enrollees but do not lower point-of-sale prices.¹⁴ In cases where a generic price is not discounted enough relative to the brand-name price, both plans and patients may have financial incentives to use brand-name drugs due to the manufacturer discounts that are required in the Part D coverage gap.¹⁰

Previous research has documented two concerning patterns. First, a recent analysis of generic drug tier placement by Avalere found that generic drugs offered under Medicare Part D have shifted from lower tier placement (corresponding to lower patient out-of-pocket spending) to higher tier placement.⁶ The study found that by 2019 nearly half of all generic products were placed in nongeneric tiers, which can increase patient cost sharing. While this study raised concerns about the placement of generic drugs on Part D formularies, it did not consider formulary placement for brand-name alternatives or whether plans may be incentivizing the use of brand-name drugs relative to generics.

Second, a recent peer reviewed study showed that in 2016, 72 percent of the formularies for Part D drug-coverage-only plans (that is, stand-alone Part D plans purchased by beneficiaries in traditional Medicare) placed at least one brand-name product on a lower tier than its generic equivalent, and among drugs covered by all formularies, 5 percent of brand-name products were covered on a lower tier.⁹ While this study showed that most plans have at least one brand-name drug on a lower tier than its generic alternative, it did not examine the frequency with which this occurred across all brand-name drugs for which there are generics.

The current study aims to build upon prior work and inform policy discussions related to Part D coverage by quantifying how often brand-name drugs receive favorable formulary inclusion or placement relative to generics. Given the overall lower cost of generics relative

to brand-name drugs and the resulting savings that generics typically offer for health plans and patients, we expected plans to be more likely to offer generics or to place them on lower cost-sharing tiers when both brand-name drugs and generics are covered.

Study Data And Methods

Data Source and Product Selection

We evaluated Medicare Part D formulary coverage of generic and brand-name drugs in the period 2012–19 for all Part D stand-alone prescription drug plans and Medicare Advantage prescription drug plans. We used the Centers for Medicare and Medicaid Services (CMS) Formulary Reference File from the third quarter of each year in the study period to define a list of generic drugs for each year. Generic drugs were defined as non-brand-name therapeutic equivalent products approved through an abbreviated new drug application and were identified in our data as products without a trade name.¹⁵ In the third quarter of 2018, this file contained 6,275 drug products (including 2,615 generics) that, according to CMS, may be included on Medicare Part D plan formularies.¹⁶ Using these product lists, we identified exact brand-name–generic pairs based on the active ingredients, strengths, and dose form for each drug available in each file.¹⁷ For example, the brand-name drug Synthroid is available in twelve different doses, and each has a generic equivalent (that is, an exact match in terms of active ingredients, strength, and dose form) associated with it. Each pair is considered a unique observation in our primary analysis, meaning that the brand Synthroid and the corresponding generic levothyroxine accounted for twelve observations.

This matching process resulted in 1,361 generic drugs with an exact match to a brand-name drug in 2019. The number of pairs varied by year.

Study Measures and Analysis

For each product, we used the Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files for the last quarter of each year¹⁸ to determine median list prices for drugs whose prescriptions were filled through Part D plans in the prior quarter and the benefit design for the first quarter of the following calendar year.

First, we determined the relative frequency of three formulary coverage patterns of the brand-name–generic pairs in our analysis in each year: covering only the brand-name drug, only the generic, or both. We refer to these formulary decisions as “plan” decisions, but we recognize that pharmacy benefit managers are used by plans to make such decisions.

We constructed a measure based on the product of brand-name–generic pairs and Part D plans available in a given year to summarize coverage patterns for brand-name and generic drugs across plans. We refer to this measure as the “plan-product combination,” in which each observation represents the coverage status of one product for one plan. This measure allowed us to understand the formulary coverage patterns described above when we considered all products and plans simultaneously. In 2019 this approach resulted in 4,176,772 plan-product combinations in which plans covered either a specific brand-name drug, its generic equivalent, or both.

Second, for plans that included both a brand-name drug and its generic equivalent on their formularies in 2019, we analyzed formulary placement (whether the brand-name drug was covered on a lower tier than the generic was or vice versa, or both products were covered on the same tier) and whether plans employed utilization management restrictions (prior authorization, step therapy, or quantity limits) on brand-name or generic products to encourage or discourage utilization of either type of drug when plans covered both types. We also estimated differences in product point-of-sale prices and patient out-of-pocket spending under various coverage patterns. Point-of-sale prices in the formulary files represent the average per fill price at in-area retail pharmacies for a product and plan from the prior quarter. Patient out-of-pocket spending was estimated using the plan formulary tier and cost-sharing amounts for the initial Part D coverage phase.

Finally, we evaluated potential spending impacts on Medicare Part D related to plan preference for brand-name drugs over generics in cases where such patterns were observed. For this evaluation, we used data from the CMS Medicare Part D Drug Spending Dashboard for 2018 to assess utilization and spending on brand-name and generic products with the same active ingredients.

Limitations

Our study had some limitations. First, matching drugs by active ingredients, dose, and formulation may create redundancy by treating all doses and formulations as individual product pairs for the primary analysis. We completed two alternative analyses that were restricted to a single dose for each product, selecting the most commonly used dose and the lowest dose, respectively. In both cases our conclusions regarding patterns of brand-name and generic drug use mirrored our other findings, so the results are not shown. While prior research on brand-name–generic competition and market response within a small number of drug classes typically considered products at the molecule level, and might even have considered therapeutic substitutes when evaluating plan behavior,^{2,3,19,20} the number of products in our study precluded our use of this approach.

Second, by requiring brand-name–generic pairs to match on active ingredients, dose, and formulation, we may have excluded some products whose brand-name counterpart has exited the market. As an example, the generic product venlafaxine had five dosage formulations included in the Formulary Reference File, but none was included in our analysis because venlafaxine’s brand-name counterpart (Effexor) was withdrawn from the market around the time that an extended release formulation (brand-name Effexor Rx) was introduced. This behavior is often termed “product hopping.” Approximately half of all generic products in the Formulary Reference File were excluded for this reason, which highlights the importance of this phenomenon as an area for future work.

Third, we considered all product pairs available in a given year, regardless of the time elapsed since the generic drug’s entry into the market and the number of generic competitors or therapeutic substitutes—both of which may affect formulary coverage, placement, and prices. Because of this, year-to-year comparisons in formulary placement decisions should be viewed cautiously.

Fourth, we did not directly account for differences in plan enrollment or drug utilization in our analysis. To partly address this concern, we used Medicare claims from the most recent year available (2018) to evaluate drug utilization for products when brand-name drugs were preferred to generics in at least some cases. However, the use of products of interest might have changed between 2018 and 2019, particularly if generic uptake was delayed.

Finally, our analysis was limited to generic products with a brand-name drug that matched the generic in terms of active ingredients, dose, and formulation, so other important issues such as product hopping and delayed generic drug entry were not explored. However, we believe that comparing exact brand-name–generic pairs was the optimal approach for our study, as these products are, by definition, directly substitutable for one another. Thus, we did not evaluate the appropriateness of tiering or utilization management strategies being employed by plans for specific products, nor did we evaluate whether there were lower-cost therapeutic alternatives available when generics were placed on higher tiers than brand-name drugs were.

Study Results

For drugs with both brand-name and generic versions, most Part D plans covered the generic only (exhibit 1). In only a handful of cases did plans cover the brand-name drug only. Generic-only coverage has increased over time, with modest decreases in brand-name-only or both brand-name and generic coverage. Based on our measure of plan-product combinations, we found that less than 1 percent of our observations (0.9 percent) corresponded to coverage of brand-name products only (exhibit 2). Conversely, 84 percent of the observations corresponded to coverage of generics only, while 15 percent corresponded to coverage of both generic and brand-name versions by the same plan.

Brand-Name Drugs Covered, Generics Excluded

Of the 1,361 drug products in our analysis, there were 218 products for which at least one plan covered the brand-name while excluding the generic version in 2019. For more than half (116) of these products, this coverage pattern was used by 1 percent or less of Part D plans (online appendix exhibits 2A and 2B).²¹ For 5 products, this coverage pattern was employed by 25 percent or more of the plans.

Of the 218 products for which the brand-name version only was covered in one or more plans, most were used by very few Medicare beneficiaries. There were two notable exceptions. Colcrys, the brand-name version of colchicine (used for the treatment of gout), had 511,036 Part D claims in 2018, versus 823,633 claims for the generic product. For Colcrys, average spending per claim was \$259, versus \$214 for generic colchicine—a difference of \$45. In 2019, generic colchicine was excluded by 20–46 percent of Part D plans in favor of the brand-name product, depending on the formulation studied (data not shown).

Copaxone, the brand-name version of generic glatiramer acetate (used to treat relapsing multiple sclerosis) had 185,180 Part D claims filed in 2018, versus 27,616 for the generic product (possibly lower because it was approved only in October 2017). Average spending

per claim was \$6,524 for Copaxone and \$4,285 for generic glatiramer—a difference of \$2,239 (data not shown). In 2019, 10–13 percent of Part D plans excluded generic glatiramer in favor of the brand-name version.

In terms of expected beneficiary out-of-pocket spending for the 218 products covered as brand-name drugs only, cost sharing was typically in the form of flat copayments (78.9 percent), meaning that the underlying list price of the brand-name drug would not factor into patient out-of-pocket spending—in contrast to coinsurance, where patients pay a percentage of the drug's price. However, these brand-name products were most often placed on tier 4, which is the highest cost-sharing tier before the specialty tier and required a \$100 copayment.

Brand-Name Drugs And Generics Covered

For the 14.8 percent (618,516 of 4,176,772) of product-plan combinations where both brand-name and generic versions of the same drug product were covered on the same plan formulary (exhibit 2), we found that brand-name drugs were placed in lower cost-sharing tiers than their generic equivalents only 1 percent of the time. Generics were placed on a lower cost-sharing tier than brand-name drugs 40 percent of the time, and brand-name and generic products were placed on the same tier 59 percent of the time.

Brand-Name Drugs Placed On Lower Cost-Sharing Tiers Than Generics

There were 125 unique products in which the brand-name drug was placed on a lower formulary tier than the generic equivalent in at least one plan (9.2 percent of the 1,361 products studied) (appendix exhibits 3A and 3B).²¹ For most of the 88 products with this coverage pattern, brand-name drugs were placed on a lower cost-sharing tier than their generic equivalents in 1 percent or less of plans. In the 1 percent of cases (7,761 of 618,516 observations) in which brand-name drugs were placed on a lower tier than their generic equivalents when both products were covered in the same plan, mean list prices were \$310 for brands and \$139 for generics (exhibit 3). However, median list prices were very low—just over \$20—for both brand-name drugs and generics. Under this coverage pattern, based on list prices and cost-sharing amounts charged for these products, the expected mean out-of-pocket spending would be \$34 for brand-name drugs and \$56 for generics. The higher expected mean out-of-pocket spending for generics reflects plan use of coinsurance for generics when brand-name drugs are preferred (36 percent of plans that used this coverage strategy required coinsurance for generics, while only 4 percent required it for the brand-name equivalents).

Brand-Name Drugs And Generics On The Same Tier

There were 652 products (47.9 percent of the products in this analysis) in which brand-name drugs and their generic equivalents were placed on the same tier by at least one plan in 2019. Of these 652 products, 465 (71.3 percent) were covered on the same tier in 10 percent or less of the plans (appendix exhibits 4A and 4B).²¹

In the 59 percent (365,160 out of 618,516) of observations in which brand-name drugs and generics were both covered and on the same tier, mean list prices were \$349 for brand-name drugs and \$197 for generics (exhibit 3).

When generics and brand-name drugs were both covered on the same tier, beneficiary cost sharing took the form of flat copayments instead of coinsurance in 80 percent of the observations. This resulted in virtually no difference in expected beneficiary out-of-pocket spending between brand-name and generic drugs: expected Median out-of-pocket spending was \$26 for the former and \$23 for the latter. However, when we focused only on the products subject to coinsurance, we found that there would be greater savings for patients who chose generic drugs over brand-name ones, even when they were both covered on the same tier. This result was partially driven by the higher average price of the brand-name products: \$1,263 (standard deviation: \$2,896) for brand-name drugs versus \$638 (SD: \$1,397) for generics (data not shown). When paying coinsurance, beneficiaries would face expected average out-of-pocket spending of \$199 (SD: \$425) for generics and \$393 (SD: \$895) for brand-name drugs, resulting in expected per fill average savings of \$194 (SD: \$597) for generic drug use.

Generics Placed On Lower Cost-Sharing Tiers Than Brand-Name Drugs

There were 1,334 products (98 percent of the products in our analysis) in which generics were placed on a lower tier than their brand-name equivalents in at least one plan in 2019. Nearly all of these generic products were placed in lower cost-sharing tiers than brand-name drugs in at least half of the plans.

When generics were placed on lower cost-sharing tiers relative to their brand-name counterparts in the same plan (245,595 of 618,516 observations), mean list prices were \$106 for generics and \$406 for brand-name drugs (exhibit 3). Based on total prices and formulary cost-sharing amounts, the expected average out-of-pocket spending would be \$17 for generics and \$108 for brand-name drugs—a per fill average savings of \$91 (SD: \$169) for generic drug use.

Utilization Management Practices For Brand-Name Drugs And Generics

For brand-name–generic pairs covered in the same plan, we found that quantity limits, prior authorization, and step therapy were rarely used for either the brand-name or the generic product in 2019 (exhibit 4). Quantity limits were the most common utilization management strategy (17 percent of observations), but these access restrictions were nearly always applied to brand-name and generic products similarly.

Discussion

Despite recent reports regarding Part D plan preference for brand-name drugs over generics, we found no evidence that such behavior was widespread among the plans in 2019. Our results are consistent with those of prior research in that we observed some cases of preferential brand-name drug placement on Part D plan formularies.⁹ However, when we examined the frequency with which this occurred across all brand-name drugs for which there were generics, we found that the opposite behavior was far more common: 84 percent

of all product-plan combinations covered the generic and excluded the brand-name drug in 2019, up from 69 percent in 2012. Our findings are also consistent with those of a recent study showing that Medicare formularies tend to incentivize the use of generics over brand-name drugs²² and with prior economics literature showing robust adoption of generic drugs after patent expiration, particularly for older products.^{2–5}

Our analysis shows that it was uncommon for brand-name drugs to receive preferred placement over their generic equivalents—by plans either excluding the generic entirely or placing the brand-name drug on a lower cost-sharing tier when covering both versions. These coverage patterns were observed for a small number of products, most of which were not widely used by Medicare beneficiaries, and typically in a small number of plans. Rather, our analysis shows that the conventional coverage pattern—generics preferred (that is, covered on lower cost-sharing tiers) to brand-name drugs—reflects the norm for Part D plans in 2019 and earlier years.

While preferential formulary placement of brand-name drugs or exclusion of generic drugs was not common in our analysis, we found that a small number of brand-name drugs were subject to higher rates of preferential treatment on Part D plan formularies, despite the availability of generic equivalents. Although most of these brand-name drugs are not highly utilized and therefore are not particularly costly to Medicare or its beneficiaries, there are some exceptions. Better inclusion and coverage of these generic drugs and of all excluded or nonpreferred generic drugs on Part D formularies could present an opportunity for cost savings to the Medicare program as well as its beneficiaries.

We found no evidence of differential use of prior authorization, step therapy, or quantity limits to influence drug selection when brand-name drugs and generics were both offered on a formulary. One possible reason for such low use of these utilization management tools could be the presence of other generic substitution laws that encourage the dispensing of generics over brand-name drugs unless dispense-as-written is indicated by the prescriber. Given the large proportion of plans that elect to cover only a generic drug on their formulary, this may further reduce the need for utilization management of such products.

Finally, for the subset of observations in which generics and brand-name drugs were placed on the same tier, it is possible that plans are indifferent in their coverage strategy for brands and their generic equivalents if both have low or similar list prices, or if the plans are unable to negotiate rebates on the brand-name drugs that would shift the financial incentive to giving them preferred placement, in relation to their generic equivalents. While beyond the scope of our analysis, this behavior merits further exploration since it has cost implications for both plans and patients. Patients who used a brand-name drug instead of a generic when both products are covered on the same tier could face unnecessary financial burden when paying deductibles or coinsurance (versus flat copayments) if the underlying list price of the brand-name drug was higher than that of the generic.

Conclusion

Overall, our analysis suggests that the vast majority of Part D plan formularies are designed to encourage utilization of generic drugs rather than their brand-name counterparts, with most plans covering only generic versions of drug products or placing generics on lower cost-sharing tiers than brand-name drugs. Given that preferential tiering of brand-name products and the exclusion of generics is not widespread in Part D plans, this suggests that policy making designed to address this relatively infrequent coverage pattern may be unlikely to yield significant savings.

Policy makers should continue to monitor Part D formulary coverage patterns to ensure consistent and generous coverage for generic drugs, given their important role in reducing prescription drug spending.

Supplementary Material

Refer to Web version on PubMed Central for supplementary material.

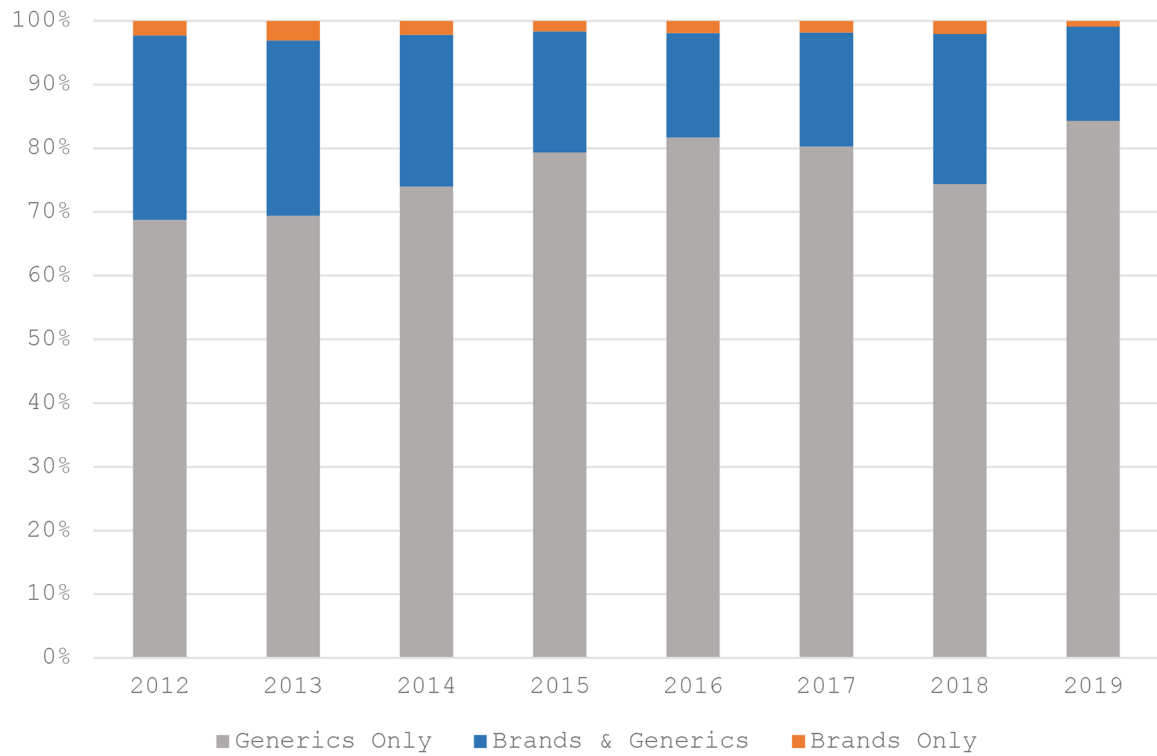
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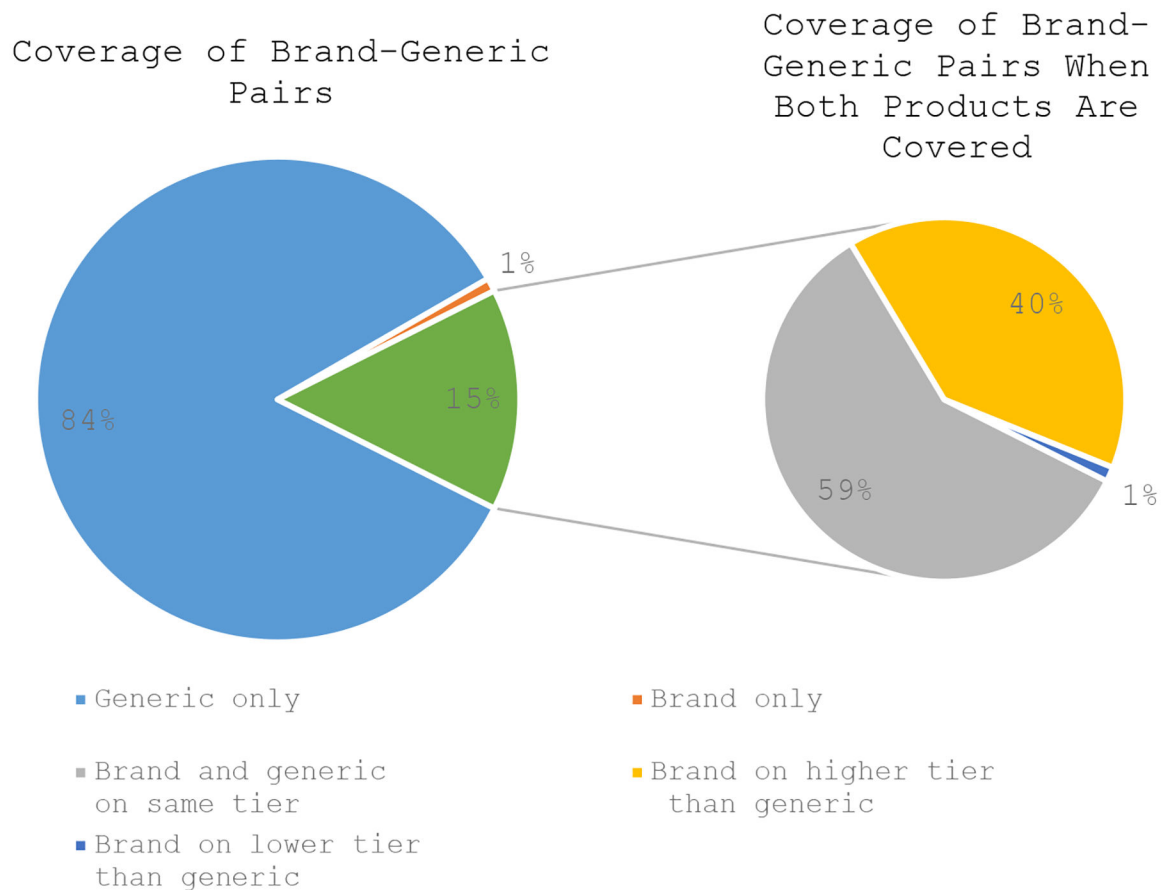
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**Exhibit 1:**

Formulary coverage among Medicare Part D plans for brand-name drugs and their generic counterparts, 2012–19

Source/Note: SOURCE Authors' analysis of data for the fourth quarter of 2018 from the Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files of the Centers for Medicare and Medicaid Services. NOTE Drugs were included in the analysis if the brand-name and generic version were exact matches based on active ingredients, strengths, and dose form.

**Exhibit 2:**

Formulary coverage of brand-name and generic pairs among Medicare Part D plans in 2019

Source/Note: SOURCE Authors' analysis of data for the fourth quarter of 2018 from the Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files of the Centers for Medicare and Medicaid Services. NOTE The unit of analysis is the product of the number of brand-name and generic drug pairs (1,361) and the number of Part D plans (3,647), plans were defined by unique contract, plan, segment, and formulary identifiers in the Medicare formulary data. "Tiers" are cost-sharing tiers.

Exhibit 3:

List prices and estimated out-of-pocket spending in the initial coverage phase for plans with brand-name drugs and their generic counterparts on the formulary

	Mean	SD	Median	25th percentile	75th percentile
Brand-name drugs preferred to generic counterparts when both covered (7,761 plan-product combinations)					
Generic list price	\$139	\$617	\$21	\$10	\$102
Brand-name list price	310	1,815	22	14	122
Generic out-of-pocket spending	56	180	37	7	90
Brand-name out-of-pocket spending	34	289	7	3	37
Brand-name drugs and generic counterparts on the same tier (365,160 plan-product combinations)					
Generic list price	197	666	44	17	132
Brand-name list price	349	1,350	52	18	208
Generic out-of-pocket spending	67	197	23	9	89
Brand-name out-of-pocket spending	104	413	26	9	95
Generic counterparts preferred to brand-name drugs when both covered (245,595 plan-product combinations)					
Generic list price	106	313	23	10	90
Brand-name list price	406	972	156	59	423
Generic out-of-pocket spending	17	34	8	2	15
Brand-name out-of-pocket spending	108	177	95	47	100

SOURCE Authors' analysis of data for the fourth quarter of 2018 from the Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files of the Centers for Medicare and Medicaid Services. NOTES The unit of analysis is the plan-product combination that represents the product of brand-name and generic pairs and Part D plans, using each strategy specified (based on the number of unique contract, plan, segment, and formulary identifiers in the Medicare formulary data), when both brand-name drugs and their generic counterparts are included on the same plan's formulary. SD is standard deviation.

Exhibit 4:

Utilization management for brand-name and generic pairs offered on the same formulary in 2019

	Generic only	Brand-name only	Both	Neither
Quantity limits	0.2%	0.2%	17.0%	82.6%
Prior authorization	0.1	2.3	9.5	88.1
Step therapy	0.0	1.5	0.7	97.8

SOURCE Authors' analysis of data for the fourth quarter of 2018 from the Prescription Drug Plan Formulary, Pharmacy Network, and Pricing Information Files of the Centers for Medicare and Medicaid Services. NOTES The exhibit shows the percentages of plan-product combinations subject to the specified utilization management options. Drugs were included in the analysis if the brand-name drug and its generic counterpart were exact matches, based on active ingredients, strengths, and dose form.